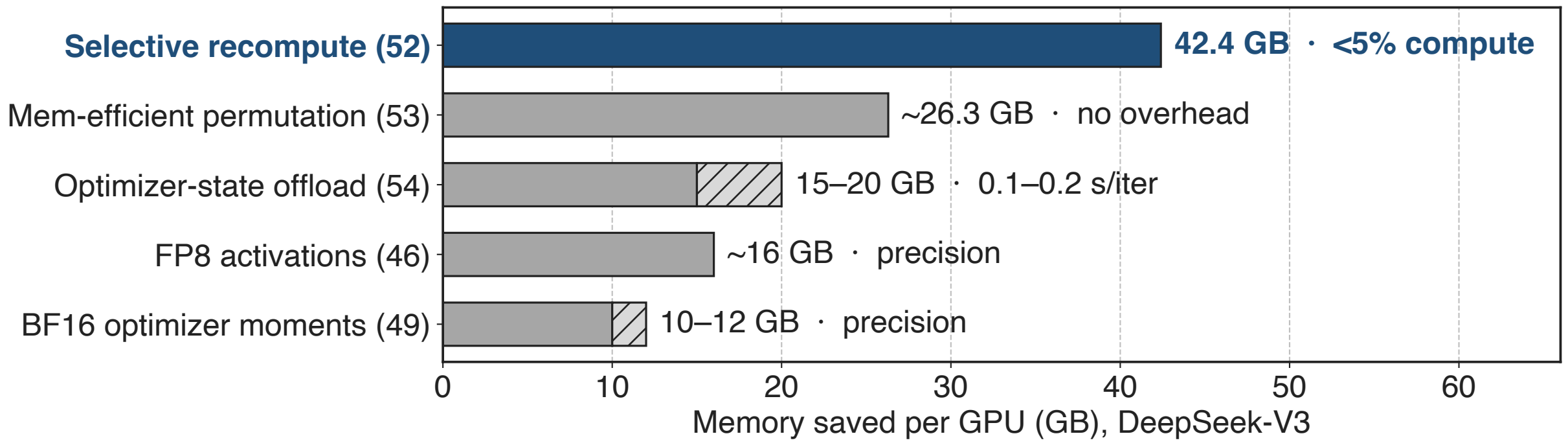
Selective recompute (52)
42.4 GB · <5% compute
Mem-efficient permutation (53)
~26.3 GB · no overhead
Optimizer-state offload (54)
15–20 GB · 0.1–0.2 s/iter
FP8 activations (46)
~16 GB · precision
BF16 optimizer moments (49)
10–12 GB · precision
0
10
20
30
40
50
60
Memory saved per GPU (GB), DeepSeek-V3